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Ans.1.
Code:

```c
#include <stdio.h>
#include <omp.h>

int main() {
    #pragma omp parallel
    {
        int thread_id = omp_get_thread_num();
        int num_threads = omp_get_num_threads();
        printf("Hello from thread %d out of %d threads\n",
               thread_id, num_threads);
    }
    return 0;
}
```

Output:

```
mnit@mnit-OptiPlex-5040:~/pdclab$ gcc -o omp_hello -fopenmp omp_hello.c
mnit@mnit-OptiPlex-5040:~/pdclab$ ./omp_hello
Hello from thread 2 out of 8 threads
Hello from thread 7 out of 8 threads
Hello from thread 1 out of 8 threads
Hello from thread 4 out of 8 threads
Hello from thread 0 out of 8 threads
Hello from thread 6 out of 8 threads
Hello from thread 5 out of 8 threads
Hello from thread 3 out of 8 threads
mnit@mnit-OptiPlex-5040:~/pdclab$
```

Ans.2.

Code:

```c
#include <stdio.h>
#include <omp.h>

#define SIZE 1000000

int main() {
    int arr[SIZE];
    long long sum = 0;

    for (int i = 0; i < SIZE; i++)
        arr[i] = i + 1;

    double start_time = omp_get_wtime();

    #pragma omp parallel for reduction(+:sum)
    for (int i = 0; i < SIZE; i++) {
        sum += arr[i];
    }

    double end_time = omp_get_wtime();

    printf("Sum: %lld\n", sum);
    printf("Time taken: %f seconds\n", end_time - start_time);

    return 0;
}
```

Output:

```
mnit@mnit-OptiPlex-5040:~/pdclab$ gcc -o omp_array_summation -fopenmp omp_array_summation.c
[1]+  Done                    gedit omp_array_summation.c
mnit@mnit-OptiPlex-5040:~/pdclab$ ./omp_array_summation
Sum: 500000500000
Time taken: 0.000682 seconds
mnit@mnit-OptiPlex-5040:~/pdclab$ export OMP_NUM_THREADS=2
mnit@mnit-OptiPlex-5040:~/pdclab$ ./omp_array_summation
Sum: 500000500000
Time taken: 0.001059 seconds
mnit@mnit-OptiPlex-5040:~/pdclab$ export OMP_NUM_THREADS=4
mnit@mnit-OptiPlex-5040:~/pdclab$ ./omp_array_summation
Sum: 500000500000
Time taken: 0.000939 seconds
mnit@mnit-OptiPlex-5040:~/pdclab$ export OMP_NUM_THREADS=6
mnit@mnit-OptiPlex-5040:~/pdclab$ ./omp_array_summation
Sum: 500000500000
Time taken: 0.000853 seconds
mnit@mnit-OptiPlex-5040:~/pdclab$ export OMP_NUM_THREADS=8
mnit@mnit-OptiPlex-5040:~/pdclab$ ./omp_array_summation
Sum: 500000500000
Time taken: 0.000746 seconds
```